

Supplementary Materials and Reproducibility Instructions

Purpose of this supplementary file

This revised supplementary file is organized to support reproducibility review. It adds a formal supplementary methods and repository guide before the existing supplementary tables. The tables remain part of the supplementary evidence, but the added sections explain how the data sources, model specifications, raw outputs, scripts, and derived tables should be stored so that reviewers can reproduce or audit the analyses.

Repository URL

Public repository for reproducibility files: <https://github.com/StevenParker-collab/ashkenazi-dna-pci-reproducibility>. The repository should be public at the time of submission and should contain a tagged or archived version matching the submitted manuscript.

Supplementary file index

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Repository structure

```
ashkenazi-dna-under-the-microscope/
  README.md
  METHODS.md
  DATA_AVAILABILITY.md
  data/
    README.md
    population_labels_qpadm.csv
    population_labels_fst.csv
    global25_coordinates_used.csv
    y_dna_frequency_sources.csv
  models/
    qpadm_model_specifications.tsv
    fst_comparisons.tsv
  scripts/
    00_environment.txt
    01_reproduce_global25_distances.py
    02_rebuild_tables.py
    03_run_qpadm_commands.sh
    04_run_fst_commands.sh
  outputs/
    raw/
    derived/
  figures/
  supplementary/
    supplementary_materials.pdf
```

Minimum data and metadata requirements

The repository should identify the public data source for each analysis, including the data provider or publication, accession or download location when available, version or access date, population labels, target labels, source labels, and any sample exclusions. If the analysis uses an external web-hosted environment, the repository should preserve exported model specifications and raw output logs so that the exact submitted results can be audited.

qpAdm reproducibility instructions

For each qpAdm model, provide a row in `models/qpadm_model_specifications.tsv` with `model_id`, target, source populations, right populations, platform or software, date run, output filename, p-value, coefficient estimates, standard errors, and feasibility status. The raw output for each run should be stored in `outputs/raw/qpadm/`. Derived tables should be generated from these outputs by `scripts/02_rebuild_tables.py`.

FST reproducibility instructions

For each FST comparison, provide a row in `models/fst_comparisons.tsv` with `population_1`, `population_2`, `comparison_set`, platform or software, output filename, FST value, standard error if available, and notes. Raw outputs should be stored in `outputs/raw/fst/`. Derived distance rankings and calibration tables should be generated from the raw table.

Global25 reproducibility instructions

Place the exact coordinate file used for the analysis in `data/global25_coordinates_used.csv`. The script `scripts/01_reproduce_global25_distances.py` calculates Euclidean distances between a target label and all comparator labels, ranks populations by distance, and writes the output to `outputs/derived/global25_distances.csv`. This procedure reproduces the Global25 distance tables and any heatmap-style rankings derived from them.

Y-DNA and literature-derived tables

For Y-DNA and literature-derived summary tables, provide `data/y_dna_frequency_sources.csv` with study, year, population, sample_size, marker, frequency, whether the value is exact or approximate, and citation. Missing values are coded as NR. This allows the supplementary tables to be rebuilt from transparent extraction records rather than retyped from the manuscript.

Reproduction checklist for reviewers

1. Clone the public repository and confirm the repository tag matches the submitted manuscript version.
2. Install the listed software environment or inspect the exported web-platform output files.
3. Verify that the public data source and population labels match the labels used in the manuscript.
4. Reproduce or audit qpAdm models using the model specification files and raw outputs.
5. Reproduce or audit pairwise FST comparisons using the comparison list and raw outputs.
6. Run the Global25 distance script and compare the ranked output with the supplementary table.
7. Run the table-building script and confirm that the generated tables match the submitted supplementary materials.
8. Check that any non-reproducible web-interface step is documented by exported outputs, access date, and settings.

Software and methods cited for reproducibility

ADMIXTOOLS and f-statistics: Patterson et al. (2012). qpAdm applications and ancient DNA model usage: Haak et al. (2015) and related published qpAdm studies cited in the manuscript. PCA and eigenanalysis: Price et al. (2006) or Patterson et al. (2006). IBD-based inference: cite the method reference used by the relevant published study. Global25 and DNAgenics are treated as descriptive or supplementary tools and should be supported by exported inputs, outputs, and settings rather than used as sole inferential evidence.

Repository version for submission

The public repository for this submission is <https://github.com/StevenParker-collab/ashkenazi-dna-pci-reproducibility>. The repository should contain the supplementary materials PDF, data-source notes, model specification files, raw exported outputs where available, and scripts or command notes used to reproduce the derived tables. A tagged release should be created when the submitted manuscript version is finalized.

Supplementary Materials

Supplementary Table S1

Table 1. Y-DNA haplogroup frequencies among Ashkenazi Jews (E, J1, J2 only).

Study	Ashkenazi Study sample size	E- E total	E- M78	E- M123	J1- M267	J2- M172	Notes
Hammer et al. 2009	1,575 (pooled)	NR	~3	~17	~17	~20	Ashkenazi and non-Ashkenazi Jews pooled; subgroup frequencies not reported
Behar et al. 2004	~442	~16.1	NR	NR	~19	~19	No Southern European comparators; coarse resolution
Semino et al. 2004	77	NR	5.2	11.7	14.6	23.2	Only study including Southern Italians; higher subclade resolution
Nebel et al. 2001	~79	23	NR	NR	~19	~24	No Southern European comparators
Shen et al. 2004	~20	NR	~10	~10	~20	~15	Very small sample; limited resolution

Table 1 note. In all tables in this manuscript, missing source values are coded as NR (not reported in the cited study). Values prefixed by ~ are approximate as reported. All haplogroup frequencies are percent.

Supplementary Table S2

Table 2. Semino et al. (2004) Y-DNA haplogroup frequencies for Ashkenazi Jews and Southern Italian regions (E, J1, J2 only).

Population	Sample size	E-M78	E-M123	J1-M267	J2-M172
Ashkenazi Jews	77	5.2	11.7	14.6	23.2
Italian (Apulia)	86	0.0	11.6	2.3	29.1
Italian (Calabria 1)	80	16.3	2.5	1.8	22.8
Italian (Calabria 2, Albanian community, Cosenza)	68	5.9	13.2	0.0	20.0
Italian (Sicily)	55	12.7	3.6	7.1	16.7

In contrast, Levantine populations are characterized by a fundamentally different Y-chromosomal structure, with J1 typically exceeding J2 and E-M123 occurring at substantially lower frequencies than those observed in both Ashkenazi Jews and Southern Italians. This inversion of haplogroup dominance is incompatible with a Levantine paternal majority model for Ashkenazim and instead supports a Mediterranean paternal structure shaped by founder effects and long-term endogamy.

Supplementary Table S3

Table 3. Y-DNA haplogroup structure in Ashkenazi Jews compared with Levantine populations (E, J1, J2 only).

Population	E-M78	E-M123	J1-M267	J2-M172
Ashkenazi Jews	5.2	11.7	14.6	23.2
Lebanese	~2 to 5	~4 to 5	~25 to 35	~10 to 15
Palestinians	~1 to 3	~2 to 4	~35 to 45	~10 to 15
Iraqis	~1 to 3	~2 to 3	~40 to 50	~10 to 15

Note. Values for Levantine and Iraqi populations are approximate ranges synthesized from Hammer et al. (2009), Behar et al. (2010), Nebel et al. (2001), and relevant studies cited in the text. All haplogroup frequencies are percent.

Supplementary Table S4

Table 4. Upper-range regional J1-M267 frequencies in Southern Italy and the Aegean.

Region	Population	J1-M267 (%)
Italy	Sicily, Agrigento	11.1
Italy	Western Sicily	10.13
Greece	Crete, Nea Nikomedeia	10.5
Italy	Trapani, Sicily	8.82
Italy	Campania, Benevento	8.3
Greece	Crete	8.3
Italy	Sicily, Southwest	7.0
Italy	Calabria, Tyrrhenian Calabria	6.8

Frequencies shown represent high reported regional J1-M267 values in Southern Italy and the Aegean. Agrigento and other synthesized values are reported in Sarno et al. (2017) and derive from underlying province-level datasets, including Boattini et al. (2013). Western Sicily and Trapani values are taken directly from Boattini et al. (2013). The Nea Nikomedeia value is reported in King et al. (2008). All listed populations retain a J2-dominant Mediterranean paternal structure.

Supplementary Table S5

Table 5. Convergent autosomal estimates of Southern Italian and central Mediterranean ancestry in Ashkenazi Jews.

Study	Methodology	Data type	Southern Italian / Mediterranean component	Additional components (Local ancestry)
Lerga-Jaso et al. (2025)	IBD-based inference (Orchestra; plus deep-learning smoothing)	Modern genomes	~68% Italian	~16.6% Levantine; ~7.2% Iraqi-Iranian-Caucasian-Turkish; ~2.4% Greek-Balkan; ~1.7% Eastern European
Waldman et al. (2022)	qpAdm (historically constrained model)	Modern genomes	~65% Southern Italian	~19% Lebanese; ~16% Eastern European
Brace et al. (2022)	qpAdm anchored to Medieval Jewish; medieval ancient DNA (Chapelfield) plus modern	Medieval ancient DNA plus modern	~67% Sicilian (Southern Italian proxy)	~33% Turkish Jewish; ~0% French; ~0% Polish

Table 5 note. Components are reported as proportions of modeled ancestry (percent). Point estimates differ because methods partition ancestry differently and qpAdm results depend on the proxy populations available. In particular, Lebanese populations are eastern Mediterranean and shifted toward Southern European genetic space relative to Arabian-associated Levantine poles; therefore, a Lebanese component in qpAdm should not be equated with a Levantine-majority origin claim. Across all approaches, Southern Italy remains the dominant autosomal source, with additional components reflecting broader eastern Mediterranean continuity and later Eastern European admixture.

Supplementary Table S6

Table 6. Autosomal F_{ST} proximity of Ashkenazi Jews in two independent studies.

A. Tian et al. (2009): SNP-based autosomal F_{ST}		B. Zoossman-Diskin (2010): autosomal genetic distance matrix ($\times 1000$)	
Comparison population	F_{ST}	Comparison population	Distance ($\times 1000$)
Italians	0.0040	Italians	44
Greeks	0.0042	Greeks	105
Spanish	0.0056	Turks	170
Tuscans	0.0066	Germans	131
Germans	0.0072	French	144
Druze	0.0088	British	238
Palestinians	0.0093	Poles	195
Irish	0.0109	Russians	230
Swedes	0.0120	Palestinians	277
Russians	0.0137		
Basque	0.0144		

Table 6 note. Use NR for values not reported. Use ~ for approximate values. Both studies independently identify Italians and Greeks as the closest populations to Ashkenazi Jews, with Palestinians substantially more distant. Ashkenazi–Southern European distances fall in the ~0.004 to ~0.005 F_{ST} range typical of closely related regional populations, whereas Ashkenazi–Palestinian distances approach or exceed ~0.009, indicating markedly deeper divergence under these panels. These results contradict claims that Ashkenazi Jews occupy a genetic midpoint between Europe and the Levant; apparent intermediacy arises when Southern Europe is excluded or replaced with genetically inappropriate proxies.

Supplementary Table S7

Table 7. CCR5- Δ 32 allele frequencies in Ashkenazi Jews and comparative populations.

Population	Sample size	CCR5- Δ 32 allele frequency
Swedish	131	0.137
Estonian	158	0.133
Polish	30	0.133
French	230	0.089
Italian	172	0.055
Greek	160	0.044
Ashkenazi Jews	503	0.097
Lebanese	51	0.000
Saudi Arabian	100	0.000
Chinese	40	0.000
African populations	various	0.000

Table 7 note. Allele frequencies are reported as proportions (0 to 1) as presented in Stephens et al. (1998). CCR5 Δ 32 frequencies follow a European north-south cline. Ashkenazi Jews fall within the European distribution and contrast with Levantine and Arabian comparators reported as 0.000 in this dataset.

Supplementary Table S8

Table 8. Top 30 closest modern populations to Ashkenazi_Germany in modern G25 similarity mapping.

#	Population	Distance
1	Maltese	0.0180
2	Italian Calabria	0.0182
3	Italian Campania	0.0207
4	Italian Campania Naples (Campanian)	0.0224
5	Sicilian Syracuse	0.0236
6	Sicilian Trapani	0.0237
7	Greek Crete	0.0245
8	Sicilian Central	0.0245
9	Sicilian East	0.0245
10	Italian Calabria (Cosentian)	0.0248
11	Greek Crete Lasithi	0.0251
12	Italian Basilicata (Lucanian)	0.0253
13	Italian Basilicata	0.0253
14	Greek Crete Rethymno	0.0261
15	Sicilian	0.0270
16	Greek Dodecanese Kos	0.0273
17	Greek Kos	0.0273
18	Italian Apulia	0.0278
19	Greek Dodecanese	0.0280
20	Sicilian West	0.0288
21	Italian Apulia (Apulian)	0.0290
22	Greek Cyclades Amorgos	0.0298
23	Greek Crete Chania	0.0307
24	Italian Abruzzo	0.0313
25	Italian Abruzzo (Abruzzese)	0.0313
26	Greek Euboea Central	0.0318
27	Greek Crete Heraklion	0.0321
28	Greek Cyclades Milos	0.0334
29	Italian Campania Salerno (Campanian)	0.0353
30	Italian Molise	0.0353

Supplementary Table S9

Table 9. qpAdm results for Jew_Ashkenazi.HO (author-generated).

Model components	Italian component (%)	Lebanese (%)	Russian (%)	<i>p</i> value	Feasible
Italian_South.HO + Lebanese.HO + Russian.DG	69.97	22.06	7.97	0.1445	Yes
Italian_North.HO + Lebanese.HO + Russian.DG	54.53	42.57	2.90	0.0256	No
Italian_Central.HO + Lebanese.HO + Russian.DG	61.33	33.83	4.84	0.0514	Yes

Key Findings

- Southern Italian model is feasible ($p = 0.1445$) and identifies Italian_South.HO as the dominant source (~70%).
- Northern Italian model fails feasibility ($p = 0.0256$) and is rejected under this specification.
- Central Italian model is marginally feasible ($p = 0.0514$) and produces intermediate estimates consistent with a geographic gradient within Italy.
- Across all feasible models, Southern Italy remains the dominant autosomal contributor to Jew_Ashkenazi.HO.

Supplementary Table S10

Table 10. Autosomal F_{ST} distances to Jew_Ashkenazi.HO (author-generated).

Population	F_{ST} distance	Standard error
Cretan.DG	0.0049	0.00124
Italian_South.HO	0.0049	0.00033
Sicilian.HO	0.0052	0.00041
Greek.HO	0.0055	0.00035
Maltese.HO	0.0066	0.00045
Jew_Moroccan.HO	0.0091	0.00058
Jew_Iraqi.DG	0.0122	0.00138
Jew_Iranian.HO	0.0125	0.00050
Jew_Tunisian.HO	0.0127	0.00056
Jew_Libyan.HO	0.0132	0.00053
Jew_Georgian.HO	0.0133	0.00056
Jew_Yemenite.HO	0.0174	0.00054

The lowest distances are observed to Southern Italians and Cretans, followed closely by Sicilians and mainland Greeks. Maltese also fall within the same Mediterranean affinity range. In contrast, all non-European Jewish comparator groups are substantially more distant, with North African, Near Eastern, Caucasian, and Arabian Jewish populations showing F_{ST} values approximately two to three times higher than those observed for Southern Italian and Aegean populations. This ordering is inconsistent with a Europe-Levant midpoint model and instead indicates tight placement within a Southern European-European and Aegean continuum with clear separation from Levantine and non-European Jewish populations under these panels.

Supplementary Table S11

Table 11. F_{ST} calibration benchmarks.

Population pair	F_{ST} distance	Interpretation
Italian_South.HO vs Maltese.HO	0.0036	Closely related central Mediterranean populations
Jew_Ashkenazi.HO vs Italian_South.HO	0.0049	Strong regional affinity
IBS_CanaryIslands.DG vs Spanish.HO	0.0038	Iberian-derived population with additional North African ancestry

Genome-wide autosomal and ancient DNA studies show that modern Canary Islanders derive a strong majority of their ancestry from Iberian sources, with a minority Indigenous North African component often estimated at approximately 15 to 25 percent (Botigué et al., 2013; Rodríguez-Varela et al., 2017). Despite this admixture, Canary Islanders cluster tightly with Iberians in PCA space and show very low F_{ST} distances to Spanish populations. This provides an external calibration demonstrating that very low F_{ST} values on the order of 0.003 to 0.004 are compatible with strong shared ancestry in the presence of admixture and drift. Under this calibration, the Ashkenazi–Southern Italian F_{ST} distance of 0.0049 is consistent with strong regional affinity rather than weak relatedness or an intermediate position.

Supplementary Table S12

Table 12. Closest Global25 populations to Italian Jews.

Rank	Population	Distance	Rank	Population	Distance
1	Maltese	0.0239	16	Greek Cyprus	0.0322
2	Italian Calabria	0.0242	17	Cypriot	0.0326
3	Sicilian Central	0.0264	18	Italian Basilicata	0.0326
4	Italian Campania	0.0273	19	Italian Basilicata (Lucanian)	0.0328
5	Italian Campania Naples (Campanian)	0.0277	20	Greek Crete Rethymno	0.0331
6	Greek Dodecanese	0.0289	21	Italian Apulia	0.0341
7	Sicilian East	0.0290	22	Turkish Cyprus	0.0342
8	Sicilian Syracuse	0.0291	23	Greek Cyclades Amorgos	0.0350
9	Sicilian	0.0295	24	Sicilian West	0.0357
10	Greek Crete Lasithi	0.0303	25	Greek Euboea Central	0.0365
11	Greek Crete	0.0305	26	Italian Apulia (Apulian)	0.0368
12	Sicilian Trapani	0.0309	27	Greek Crete Heraklion	0.0381
13	Greek Dodecanese Kos	0.0310	28	Greek Crete Chania	0.0385
14	Greek Kos	0.0310	29	Italian Campania Benevento (Campanian)	0.0389
15	Italian Calabria (Cosentian)	0.0310	30	Italian Campania Salerno (Campanian)	0.0391

Table 12 note. Modern Global25 similarity mapping of Italian Jews identifies a strongly central Mediterranean affinity profile. The closest populations are Maltese, Southern Italians (Calabria, Campania, Sicily, Basilicata, and Apulia), Aegean Greeks (Crete, Dodecanese, Cyclades), and Cypriots. Notably, Northern Italian populations are absent from the closest matches despite their frequent use as proxies in ancestry modeling. This pattern closely parallels the results observed for Ashkenazi_Germany and provides an independent falsification test of Northern Italian proxy models. If Ashkenazi and Italian Jewish populations shared a predominantly Northern Italian ancestral source, Northern Italian populations would be expected among the nearest modern populations. Instead, both datasets resolve within the same central Mediterranean affinity space dominated by Southern Italian, Sicilian, Maltese, and Aegean populations.

Regional Composition of Top 30 Matches					
Southern Italian	Sicilian	Greek/Aegean	Maltese	Cypriot	Northern Italian
17	6	11	1	2	0